

Automated Phishing Detection & Analysis System

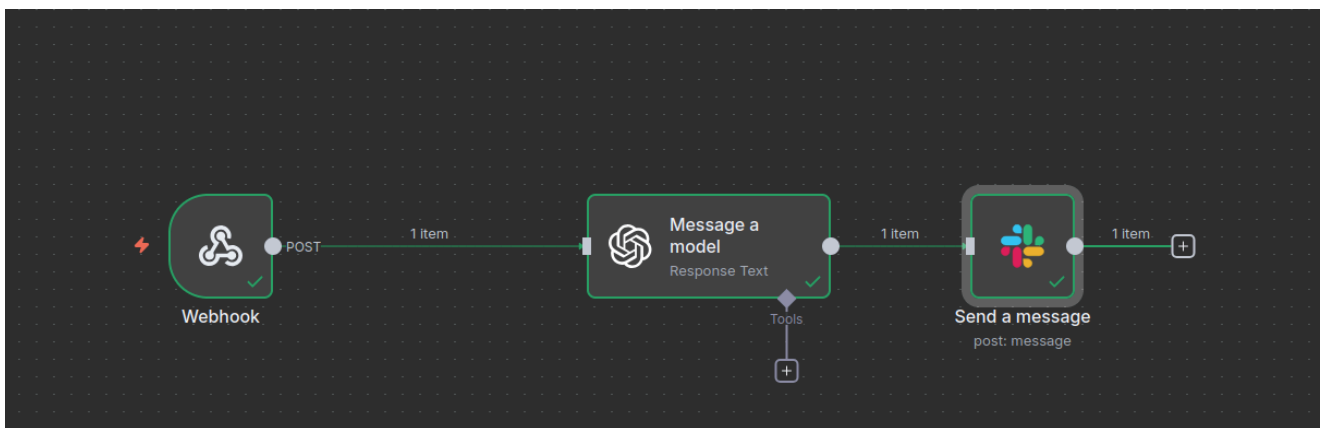
An intelligent automation workflow that integrates Splunk, n8n, and ChatGPT API to detect, analyze, and notify security teams about phishing attempts in real-time.

Overview

This project automates the detection and analysis of phishing emails using a three-stage pipeline:

1. **Detection:** Splunk monitors email traffic and identifies potential phishing attempts
2. **Analysis:** ChatGPT (GPT-4.1-MINI) performs deep analysis of suspicious emails
3. **Notification:** Slack alerts notify security teams with detailed analysis and recommendations

Architecture

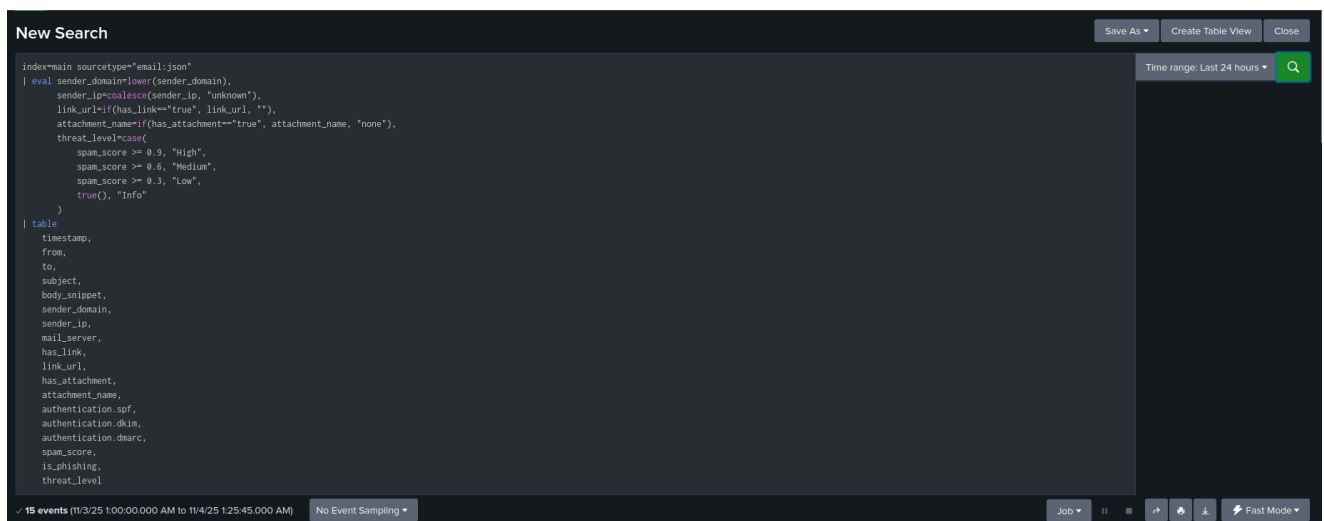


Workflow Components

1. **Splunk:** Email security monitoring and alerting
2. **n8n:** Workflow automation and orchestration
3. **ChatGPT API:** AI-powered phishing analysis
4. **Slack:** Real-time team notifications

Configuration

Splunk Alert Configuration



Webhook Configuration:

- Set webhook URL to your n8n instance: `http://[n8n-host]:5678/webhook/phishing-alert`
- Method: POST
- Format: JSON

n8n Workflow Configuration

Node 1: Webhook Trigger

- **URL Path:** `/webhook/phishing-alert`
- **Method:** POST
- **Authentication:** None

Node 2: Message a Model (ChatGPT)

- **Credential:** OpenAI account
- **Resource:** Text
- **Operation:** Message a Model
- **Model:** GPT-4.1-MINI
- **System Prompt:**

 Message a model

Execute step

ParametersSettingsDocs

Credential to connect with

OpenAi account

Resource

Text

Operation

Message a Model

Model

From list

GPT-4.1-MINI

Messages

Messages

Type

Text

Role

System

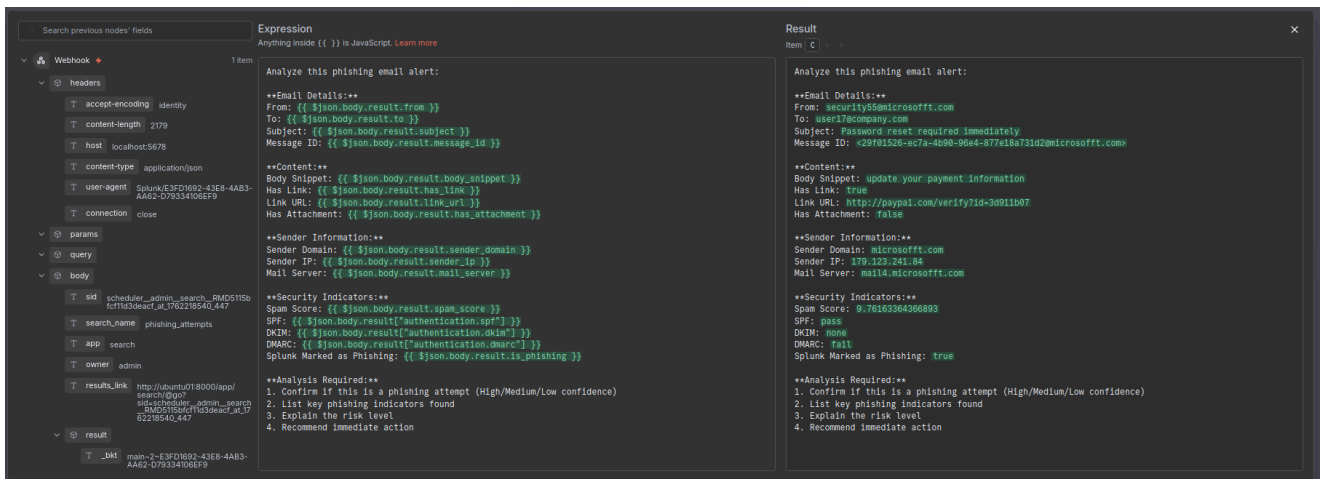
Prompt

You are a cybersecurity analyst specializing in phishing detection. You will be given email metadata, content snippets, and authentication results from a mail security system (Splunk). Analyze each email and classify it as 'Phishing' or 'Legitimate'. Explain your reasoning clearly and provide a short summary for analysts also give a recommendations.

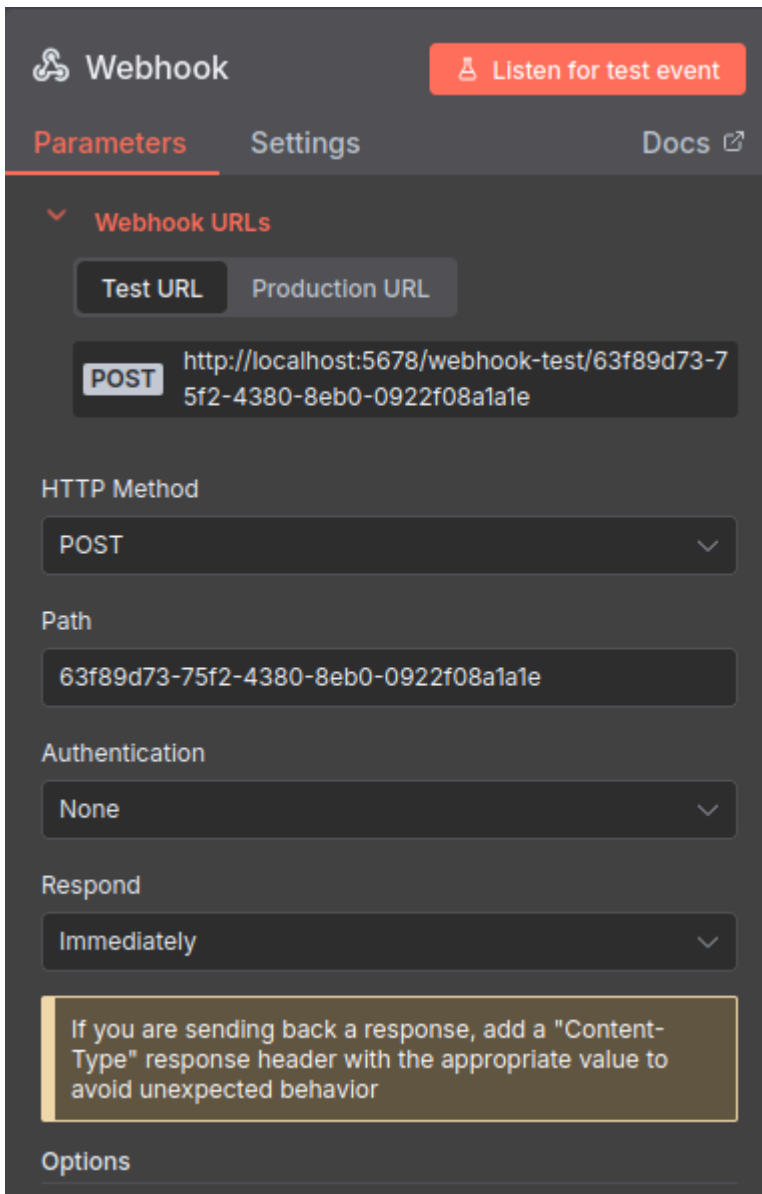
Node 3: Send Slack Message

- **Credential:** Slack account
- **Resource:** Message
- **Operation:** Send
- **Channel:** Select your security channel
- **Message Format**

The following image illustrates the Data Extraction and Analysis



Webhook Configuration as shown in the above image



Testing with slack

Send a message

⚠ Execute step

Parameters
Settings
Docs

Credential to connect with

Slack account

Resource

Message

Operation

Send

Send Message To

Channel

Channel

From list
all-soc-lab

Message Type

Simple Text Message

Message Text

/x {{ \$json.output[0].content[0].text }}

****Phishing Analysis Report** 1. **Phishing Confirmation**

No properties

Add option

OUTPUT

1 item

Schema
Table
JSON

```

[
  {
    "ok": true,
    "channel": "C09Q3B2SFJB",
    "message": {
      "user": "U090G500078",
      "type": "message",
      "ts": "1762218679.638249",
      "bot_id": "B09QJTNH22E",
      "app_id": "A09QGP30JJ",
      "text": "**Phishing Analysis Report**\n\n1. **Phishing Confirmation & Confidence Level:**\n\nThis email is a **Phishing attempt** with **High confidence**.\n\n2. **Key Phishing Indicators:**\n\n- **Suspicious sender domain:** The domain \"<http://microsofft.com|microsofft.com>\" is a common misspelling of the legitimate \"<http://microsoft.com|microsoft.com>\" which is a well-known phishing tactic (typosquatting).\n\n- **Mismatch between sender domain and link URL:** The email claims to be from \"<http://microsofft.com|microsoft.com>\" but includes a link to \"<http://paypal.com|paypal.com>\" (a misspelling of <http://paypal.com|paypal.com>), indicating a likely attempt to redirect users to a fraudulent website.\n\n- **DMARC failure:** The email failed DMARC validation which confirms the sender is likely not authorized to send as this domain.\n\n- **High spam score:** A spam score of -9.76 (out of 10) strongly indicates the message is suspicious or unwanted.\n\n- **DKIM missing:** No DKIM signature is present to verify the sender's identity authentically.\n\n- **Urgent and alarming subject/content:** Use of urgency (\"Password reset required immediately\") to prompt quick action, typical in phishing.\n\n- **Generic content snippet unrelated to sender:** \"update your payment information\" is an odd request in an email supposedly related to password reset.\n\n3. **Risk Level Explanation:**\n\n- **High Risk:** This email attempts to trick recipients into clicking a link leading to a fraudulent site, likely designed to steal login credentials or payment details. Falling for this could lead to credential compromise, financial loss, and potential breach of corporate systems.\n\n4. **Immediate Actions Recommended:**\n\n- Quarantine or block the email to prevent user access.\n\n- Alert the user targeted and educate on spotting phishing emails.\n\n- Add

```

Slack Notification Example

 Project App 1:03 AM

Analysis:

1. Phishing Confirmation: High Confidence

2. Key Phishing Indicators:

- Sender Domain Spoofing: The sender domain is microsofft.com which mimics the legitimate microsoft.com with a deliberate typo (ff instead of t).
- Suspicious URL: The embedded link points to http://paypal.com/verify?id=3d911b07 which is a lookalike of paypal.com replacing l with 1. This is a common phishing tactic to trick users.
- DMARC Fail: Despite SPF passing, DMARC fails likely due to domain spoofing, indicating mails aren't authorized by the legitimate domain owner.
- High Spam Score: The spam score is very high (9.76), which strongly correlates with phishing or spam emails.
- Mismatch Between Content and Sender: The email claims password reset required immediately but the content snippet mentions update your payment information, which is inconsistent and suspicious.
- No DKIM Signature: Absence of DKIM reduces the likelihood of authentication from a legitimate sender.
- Splunk Marked as Phishing: An internal security tool (Splunk) has flagged it as phishing.

3. Risk Level: High

- Users clicking the link may be redirected to a malicious site designed to steal login credentials or payment information.
- This can lead to account compromise, financial loss, or further internal network security breaches.
- The spoofed domain can also harm brand reputation and cause audit alarms.

4. Recommended Immediate Action:

- Block the sender domain microsofft.com and the IP 179.123.241.84 at the mail gateway.
- Quarantine or delete all emails with similar Subject or From pattern.
- Notify all users about this phishing attempt and advise not to click on the links or reply.
- Review email filtering rules to better capture lookalike domains and suspicious URLs.
- Report the malicious domain and IP to authorities and hosting providers.
- Conduct a phishing awareness training refresher focusing on domain spoofing and URL checking.

Summary for Analysts:

This email is a high-confidence phishing attempt using domain spoofing (microsofft.com), misleading URL (paypal.com), and mismatched content to trick users into divulging credentials or payment info. The email fails DMARC, has a very high spam score, and is flagged by internal tools. Immediate blocking, user awareness, and strengthened email defenses are critical to mitigate this.

Automated with this n8n workflow

 PayPal

Workflow Details

Data Flow

1. Splunk Detection Phase:

- Monitors incoming emails
- Analyzes SPF, DKIM, DMARC authentication
- Checks spam scores and suspicious patterns
- Triggers webhook when phishing detected

2. n8n Processing Phase:

- Receives webhook from Splunk
- Extracts email metadata:
 - Sender domain and IP

- Subject line
- Body snippet
- Link URLs
- Authentication results
- Security indicators

3. ChatGPT Analysis Phase:

- Analyzes email characteristics
- Identifies phishing indicators:
 - Domain spoofing
 - Suspicious URLs
 - DMARC failures
 - High spam scores
 - Content mismatches
- Provides confidence level assessment
- Generates actionable recommendations

4. Slack Notification Phase:

- Formats analysis results
- Sends to designated security channel
- Includes all relevant details for incident response



Future Enhancements

- ☐ Add automated response actions (quarantine emails)
- ☐ Implement machine learning model for local analysis
- ☐ Create dashboard for phishing trends



Acknowledgments

- Splunk for robust security monitoring capabilities
- n8n for flexible workflow automation
- OpenAI for powerful AI analysis capabilities
- Slack for seamless team communication